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Flows & Liquidity

A \$100bn rebalancing flow boosted April's equity rally

- Retail investors propelled the equity market in April via equity funds. If one includes the \$100bn estimated rebalancing flow from leveraged equity ETFs, the net flow emanating from equity funds globally reached \$190bn in April, a new record high.
- Share buybacks by US tech companies added further fuel to the equity rally in April.
- We believe that institutional investors and in particular macro managers have room to further increase their equity exposures from here.
- Strong CTA performance in April largely due to a faster shift to equity longs than after the 'Liberation Day' announcement.
- The debasement trade rotates from gold to bitcoin.
- MicroStrategy's bitcoin purchases accelerated in April.

Global Markets Strategy

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Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 29th Apr 26.

MF & ETF Flows	Min	Max	4 wk avg (\$bn)	2025 avg (\$bn)
All Equities			20.5	8.1
All Bonds			9.9	11.3
US Equities			13.4	3.5
US Bonds			2.7	4.3
Non-US Equities			7.1	4.6
Non-US Bonds			7.3	7.0
US HG Bonds			4.5	3.4
US HY Bonds			1.1	0.4
US Lev. Loans *			-0.8	0.0
US MMFs			-0.1	4.0
EM Equities			1.7	0.9
EM Bonds			1.52	0.50
Japan Equities			-0.1	-0.1
China Equities			0.09	-0.15
Europe				
Europe Equities			-1.0	0.9
Europe Bonds			3.1	4.3
Europe HG Bonds			-0.5	0.9
Europe HY Bonds			0.19	0.17
Europe MMFs			4.5	0.0
Other Equities			6.59	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

As of 05-May-26	MIN	MAX	Current percentile
Equities			0.73
Govt Bonds			0.55
Credit			0.25
Dollar			0.49
Commodities ex Gold			0.87
Gold			0.54
Bitcoin			0.66
EM Equities			0.62
EM Bonds/FX			0.36
Japan Equities			0.88
Europe Equities			0.76
US Equity Sectors:			
Energy			0.80
Materials			0.62
Industrials			0.44
Discretionary			0.61
Staples			0.20
Health Care			0.37
Financials			0.29
Technology			0.77
Communication Services			0.57
Utilities			0.47

Source: J.P. Morgan Flows & Liquidity.

Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

- Hedge funds made further gains in April led by equity focused funds such as equity long /short and technology sector hedge funds, both of which gained close to 7% in April, as well as momentum driven CTAs which gained another 2.5% in April bringing their YTD total return to close to 9% (Figure 1). On our calculations, the equity exposures of momentum-driven systematic funds such as CTA have risen further and are now in overbought territory as discussed in the section below.

Figure 1: Provisional Hedge Fund performance for April 2026 by category

Pivotal Path hedge fund return indices.

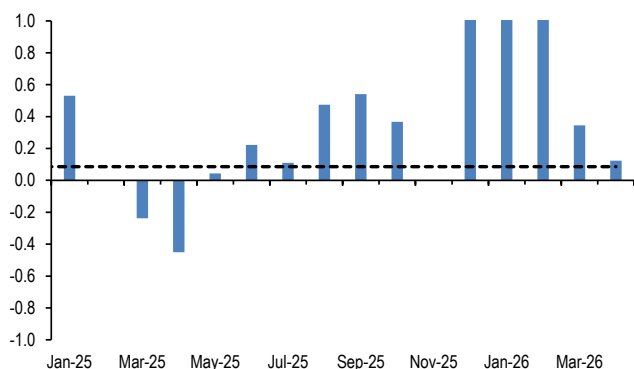
	Jan-26	Feb-26	Jan-Feb 26	Mar-26	Apr-26
ALL HFs	1.99%	1.25%	3.26%	-2.51%	5.97%
Equity L/S	2.18%	1.30%	3.51%	-4.58%	7.67%
Macro	3.20%	1.77%	5.03%	-2.48%	1.25%
Managed Futures/CTAs	3.71%	3.11%	6.94%	-0.82%	2.53%
Multi Strat	2.03%	1.13%	3.18%	-1.42%	3.37%
Credit	1.04%	0.26%	1.30%	-0.41%	0.82%
Equity sector	1.36%	1.29%	2.67%	-2.33%	6.92%
Event Driven	1.51%	0.86%	2.38%	-2.44%	7.42%
Equity Quant	-0.66%	0.95%	0.28%	0.43%	0.90%
Volatility	1.30%	1.29%	2.61%	1.23%	

Source: Pivotal Path, J.P. Morgan Flows & Liquidity.

- However, other hedge fund categories appear to still have room to increase their equity exposures.
- Macro hedge funds made only 1.2% in April. In other words, the equity betas of Macro hedge funds remained low in April suggesting that these funds have room to rebuild their equity exposures (Figure 2). The cautious stance of macro managers is also reflected in the still elevated short interest of the SPY and QQQ ETFs (Figure 3).

Figure 2: Equity beta of monthly reporting Macro hedge funds

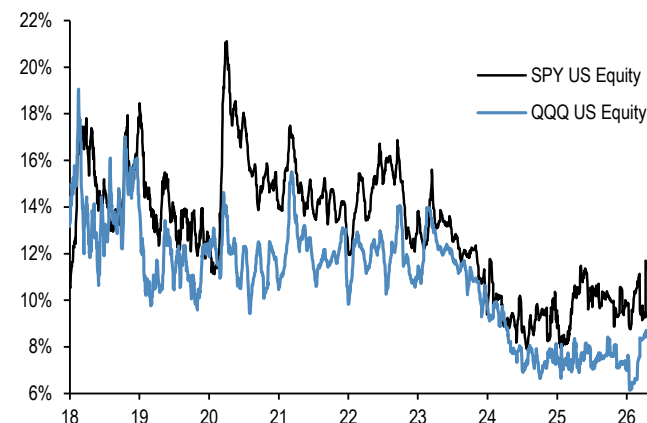
Proxied by the ratio of the monthly performance of Pivotal Path Macro hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. as of Apr 26 based on provisional data from Pivotal Path.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Figure 3: Short interest on the SPY and QQQ US ETF

Short Interest as a % share of shares outstanding. Last obs is for 5th May 2026.

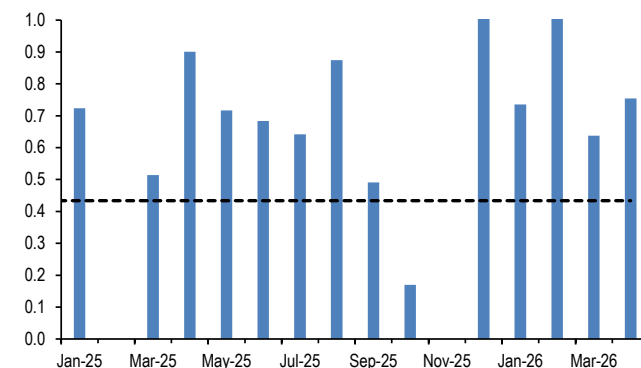


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- The equity betas of equity long/short hedge funds and TMT sector hedge funds stood above historical averages in April, but remain below recent peaks as shown by Figure 4 and Figure 5 respectively. In other words, equity long/short and TMT sector hedge funds do not appear to be fully invested yet and thus have some room to further increase their equity exposures.

Figure 4: Equity beta of monthly reporting Equity Diversified Long/Short hedge funds

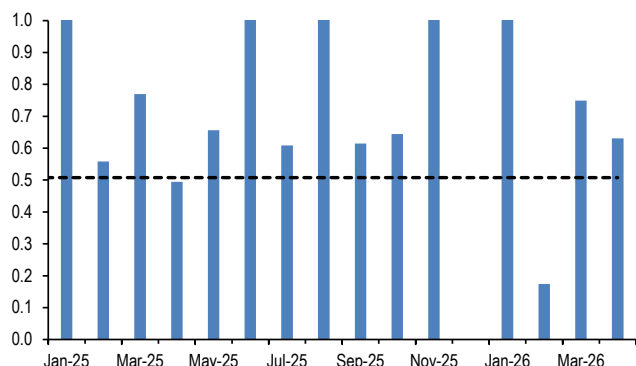
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. as of Apr 26 based on provisional data from Pivotal Path.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Figure 5: Equity beta of monthly reporting Equity TMT Sector hedge funds

Proxied by the ratio of the monthly performance of Pivotal Path equally-weighted Equity Sector Technology/Media/Telecomm hedge fund index divided by the monthly performance of Nasdaq Composite Index. Latest obs. as of Apr 26 based on provisional data from Pivotal Path.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

- Our broader equity positioning metric regularly shown in our cross-asset monitor on the front page and also shown in Figure 6 for convenience, also implies some further room for institutional investors to increase their equity exposures as this metric has yet to reach last January's high.

Figure 6: Broader equity positioning proxy from our cross-asset monitor

Percentile of its history since 2015.



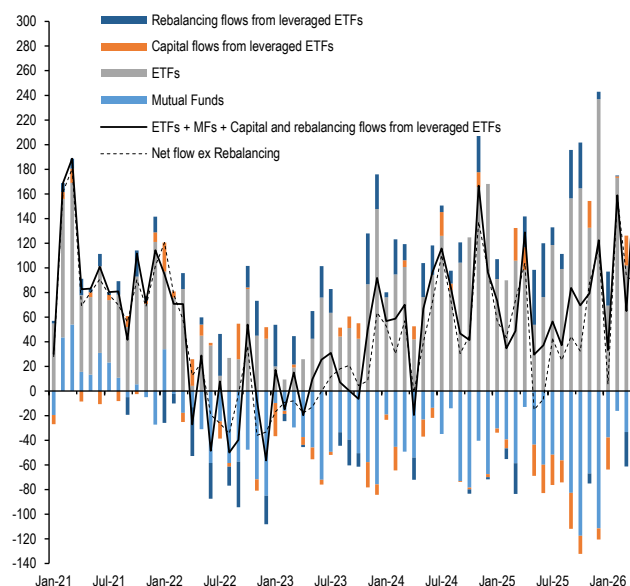
Source: J.P. Morgan Flows & Liquidity.

- What about retail investors? Retail investors' impulse proxied by equity fund flows including both mutual funds and ETFs was steady in April at around \$90bn (dotted line in Figure 7). However, if one includes the \$100bn estimated rebalancing flow from leveraged equity ETFs, the net flow emanating from equity funds globally reached \$190bn in April, surpassing the previous record high of March 2021. The \$100bn estimated rebalancing

flow by leveraged equity ETFs during April was the highest monthly rebalancing flow on record and given this flow tends to reverberate at the end of the day in a price insensitive manner, it has certainly acted as a strong amplification force for the equity market during April.

Figure 7: Equity fund flows globally by month

\$bn per month. Dotted line excludes the rebalancing flow from leveraged equity ETFs.

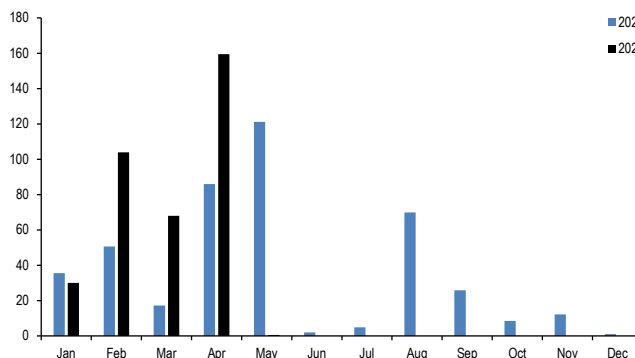


Source: LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Corporate buybacks added further fuel to the tech rally in April. As shown in Figure 8, share buybacks by US tech sector companies are tracking a YTD pace that is significantly stronger than last year.

Figure 8: US Tech sector share buyback announcements YTD vs 2025

In \$bn per month.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In all, retail investors propelled the equity market in April via equity funds. If one includes the \$100bn estimated

rebalancing flow from leveraged equity ETFs, the net flow emanating from equity funds globally reached \$190bn in April, a new record high. Share buybacks by US tech companies added further fuel to the equity rally in April. We believe that institutional investors and in particular macro managers have room to further increase their equity exposures from here.

Strong CTA performance in April largely due to a faster shift to equity longs than after the ‘Liberation Day’ announcement

- As discussed above, CTAs were among the hedge funds that have performed well over the past month. Indeed, if we look at the cumulative performance since the start of the year, the performance in April effectively recovered the drawdowns in March. What accounts for this faster recovery than we saw in the aftermath of the “Liberation Day” announcement in 2015? We look at our positioning proxies for momentum-based investors such as CTAs below.

Figure 9: HFRX Macro Systematic Diversified CTA Index



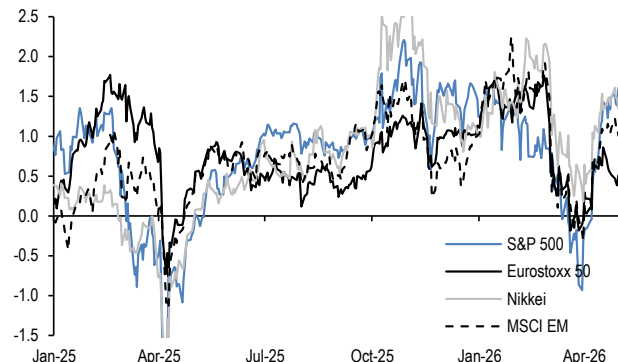
Source: HFR, J.P. Morgan Flows & Liquidity.

- Starting with equities, Figure 10 shows the average of the shorter- and longer-term signals in our positioning framework for momentum investors (see Table A3 and Table A5 in the Appendix) for US, Euro area, Japanese and EM equities. The z-score we calculate in the Appendix tables are on a very long history for the standard deviation in the numerator, while many momentum-based investors scale position limits by more recent market volatility assumptions. To adjust for this, we scale the z-score with the ratio of volatilities of the past 10 years and the past 6 months, shown in Figure 10.
- Figure 10 suggests that by the time the broader correction in equities started as the conflict in the Middle East

began, momentum-based investors such as CTAs had already trimmed positions in US equities from very elevated levels in early January. But positions in non-US equities were very elevated, contributing to the early March drawdown in CTA returns. But compared to Mar/Apr 2025, the signals only turned neutral for non-US equities, and only briefly short for US equities, before they shifted long by early April and thus took part in the rally more quickly than last year. Currently, the positions are already in rather elevated territory for US and Japanese equities with EM equities close as well, while the signals for Euro area equities are only modestly long.

Figure 10: Momentum signals for S&P 500, Eurostoxx 50, Nikkei and MSCI EM futures

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.

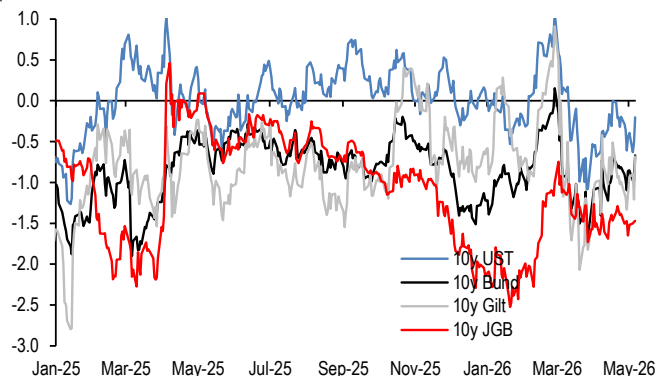


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Figure 11 shows the same adjusted z-scores for our momentum signals for 10y USTs, Bunds, gilts and JGBs. It suggests that CTAs had turned modestly long duration in USTs and gilts as the conflict began, were neutral in Bunds and still short in JGBs. These positions shifted to shorts relatively quickly, but the relatively range-bound yields for much of April suggests these positions have not contributed significantly to the recovery in CTA performance.

Figure 11: Momentum signals for 10y UST, Bund, gilt and JGB futures

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.

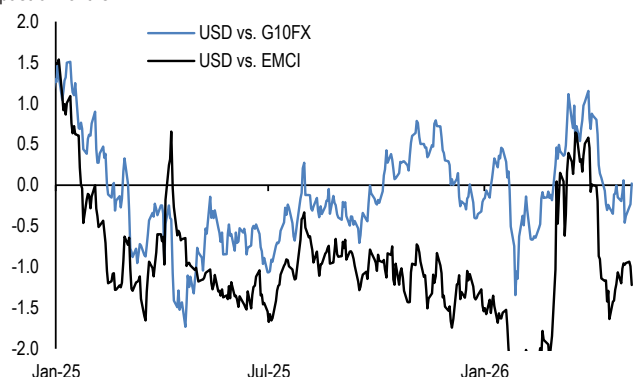


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What about currencies? Figure 12 shows the adjusted average z-score for the US dollar vs. G10 FX and EMCI. It suggests elevated shorts vs. EMCI in particular by momentum investors heading into the conflict, and the unwind of these positions and a shift to modest longs amplified the dollar rally. They shifted back to dollar shorts in early April, and this participation in the rally in EM currencies from early April has likely contributed to the recovery in CTA performance since then.

Figure 12: Momentum signals for USD vs. G10FX and EMCI

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



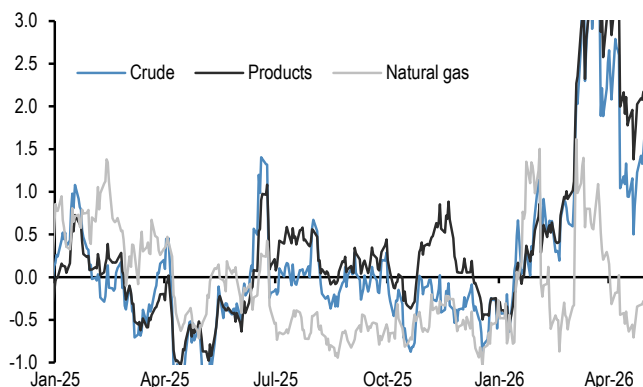
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What about commodities? In energy contracts, momentum-based investors were already somewhat long in crude (WTI and Brent) and products (RBOB gasoline, ULSD and gasoil) futures by the start of the conflict, though they were modestly short natural gas (NYM and TTF) futures. This long in crude and product futures likely helped mitigate some of the drawdown in CTA performance in

March, but continued longs in April provided little support overall to CTA performance over the month.

Figure 13: Momentum signals for energy futures (grouped into crude, products and natural gas)

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.

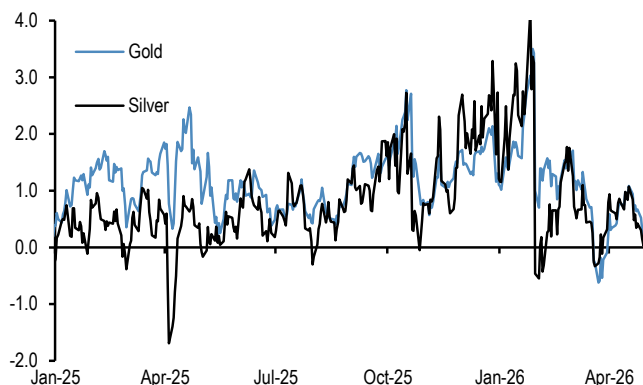


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- What about precious metals futures? Our momentum signals for gold and silver had already seen significant swings amid the late Jan correction, but momentum-based investors likely retained significant longs in end-February heading into the start of the conflict. But beyond contributing to the drawdown for CTA performance in early March, the more range-bound gold and silver prices suggest these positions played only a modest role in the April performance for CTAs.

Figure 14: Momentum signals for gold and silver futures

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

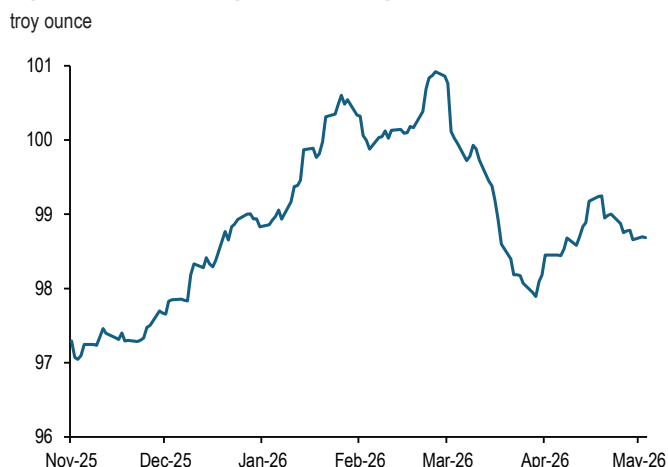
- In all, the above suggests that the positive performance of CTAs in April was likely to a significant extent due to the

relatively faster shift back to equity longs than in Mar/Apr 2025. A relatively sharp shift to short dollar vs. EM currencies also contributed, while bond and commodity positions at most made modest contributions to the positive CTA performance in April.

The debasement trade rotates from gold to bitcoin. MicroStrategy's bitcoin purchases accelerated in April

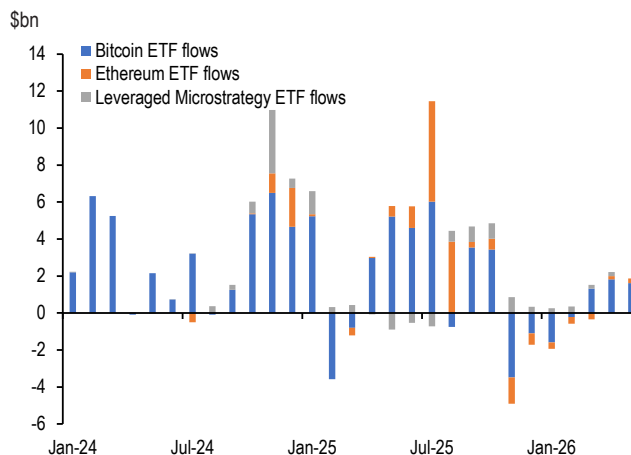
- While gold ETFs are still struggling to recapture the outflows they suffered during March as the Iran conflict erupted (Figure 15), bitcoin ETFs have been seeing inflows for a third straight month in May (Figure 16). This suggests that retail investors have shifted their preference from gold to bitcoin as an expression of the debasement trade since the start of the Iran conflict.

Figure 15: Total known gold ETF holdings



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 16: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

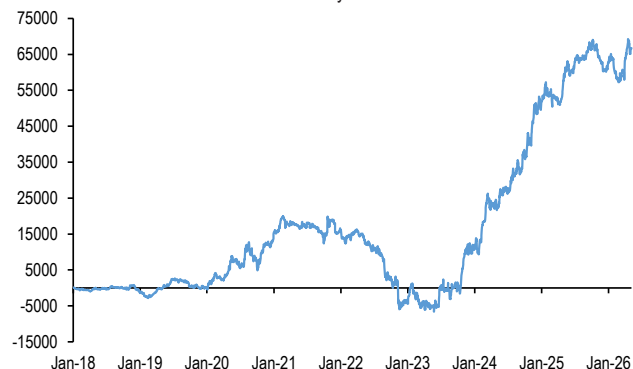


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Bitcoin buying was not only confined to retail investors but also institutional investors. Our futures positioning proxies based on CME bitcoin futures and offshore perpetual futures have been making new highs (Figure 17 and Figure 18). Our momentum signals for bitcoin and gold, as proxies for momentum traders' positioning such as CTAs, have been also rebounding for bitcoin since the start of the Iran conflict (Figure 19).

Figure 17: Our position proxy based on open interest changes in CME bitcoin futures contracts

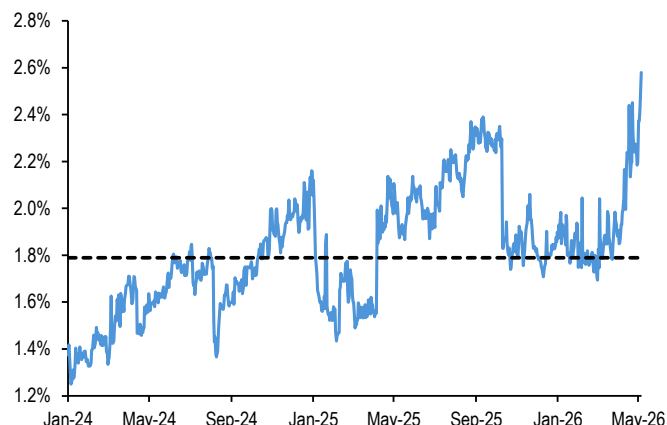
In number of contracts. Latest data until May 5th



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 18: Ratio of bitcoin perpetual futures open interest to bitcoin market cap

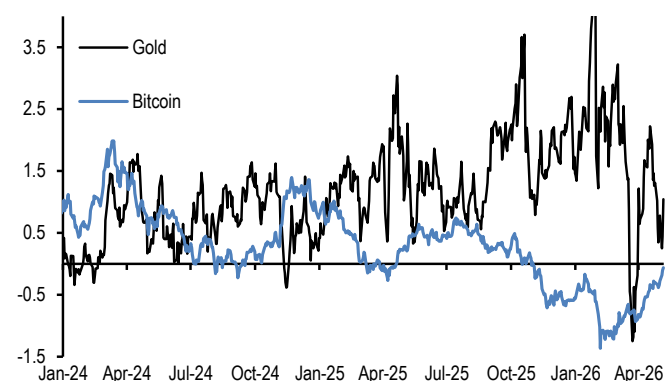
Black dotted line is the average since 2024



Source: Messari, J.P. Morgan Flows & Liquidity.

Figure 19: Our momentum signals for bitcoin and gold as proxies for momentum traders positioning

Average z-score of short- and long-term momentum signal



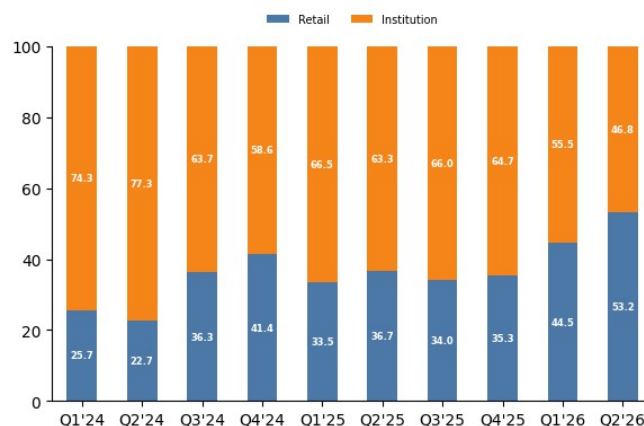
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Investors' bitcoin buying is also indirect via MicroStrategy, the ownership of which is almost equally split between retail and institutional investors (Figure 20). Because of higher investor demand, MicroStrategy's premium to NAV has expanded over the past couple of months to reach 26% currently, creating easier financial conditions and allowing the company to issue equity and debt to finance further bitcoin purchases. Indeed, MicroStrategy appears to have re-accelerated its bitcoin purchases in April, extending a 2026 pattern of increasingly opportunistic buying, responsive to both market conditions and financing availability. YTD, the company has added 145,834 BTC (roughly \$11bn), with much of that activity occurring while bitcoin traded below Strategy's estimated average purchase cost (~\$75k). On an annualized basis, the YTD pace implies roughly \$30bn of

bitcoin purchases for 2026, significantly stronger than either 2025 or 2024, with each year seeing around \$22bn of bitcoin purchases by Strategy (Figure 21).

Figure 20: Strategy's ownership type by retail and institutional

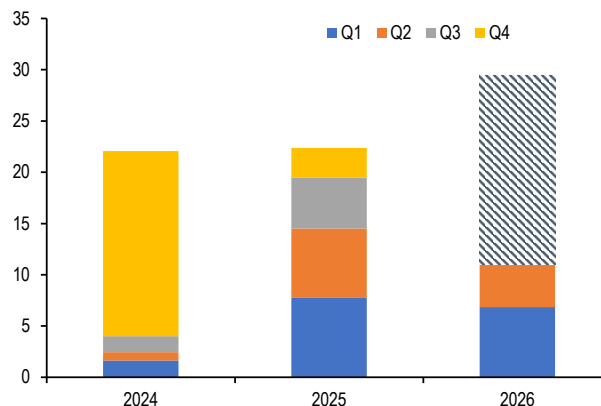
In %, bloomberg data based on quarter end dates, Q2 is QTD



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 21: MicroStrategy's bitcoins purchase each year

\$bn per year, annualised YTD pace for 2026

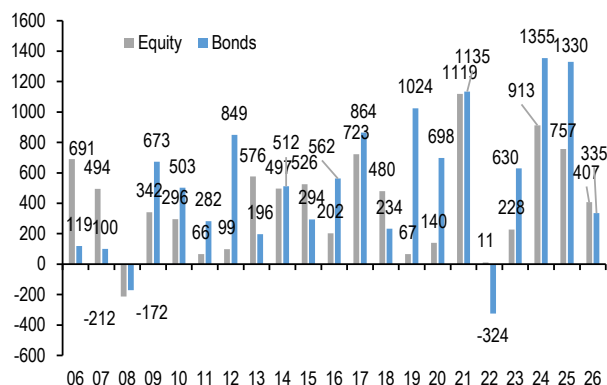


Source: Strategy filings, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Appendix

Chart A1a: Global equity & bond fund flows

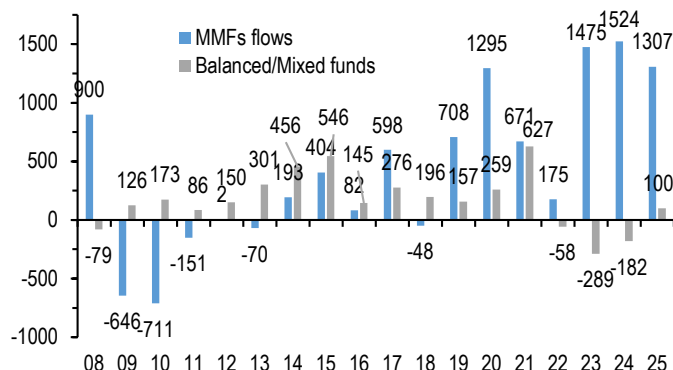
\$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q4'25). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

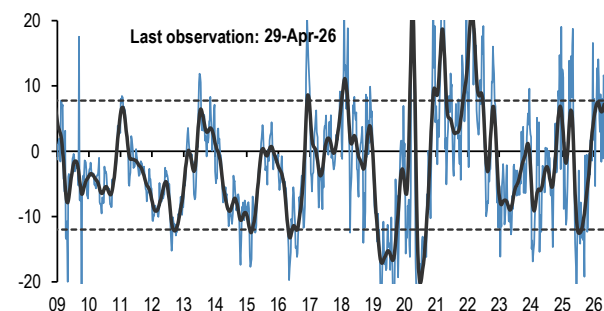
\$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q4'25.



Source: ICI, J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

Difference between flows into Equity and Bond funds: \$bn per week. Difference between flows into Equity vs. Bond funds in \$bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ± 1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Mar-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity			1.1	\$1.6	104%
DM Equity			1.5	\$14.5	52%
Govt Bonds					
UST cash			12.3	\$20.3	7%
UST futures			0.9	\$23.6	18%
JGBs			44.5	¥4,727	5%
Bund futures			1.6	€14.9	18%
Credit					
US HG			1.2	\$1.0	17%
US HY			0.6	\$0.2	3%
US Convertibles			3.8	\$0.04	18%
Commodities					
Gold			88.7	\$2.9	109%
Oil			99.7	\$6.1	-3%
Copper			3.0	\$0.9	106%
Digital Assets					
CME Bitcoin			112.3	\$0.074	-33%
CME Ethereum			103.1	\$0.033	60%

* Data with one month lag

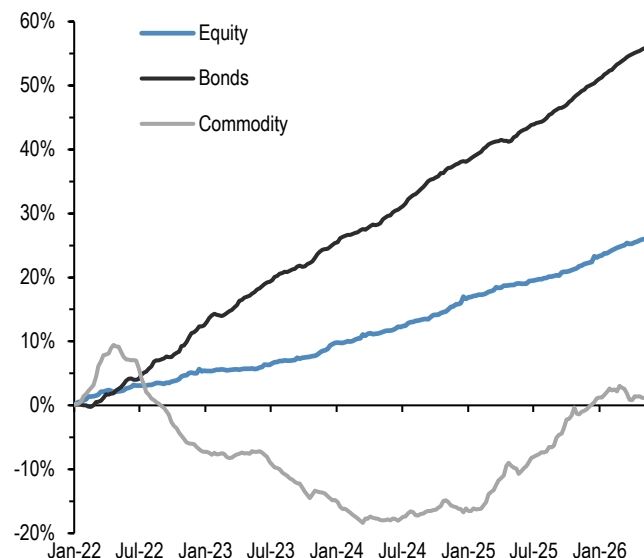
Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

06 May 2026

ETF Flow Monitor (as of 6th May)

Chart A3: Global Cross Asset ETF Flows

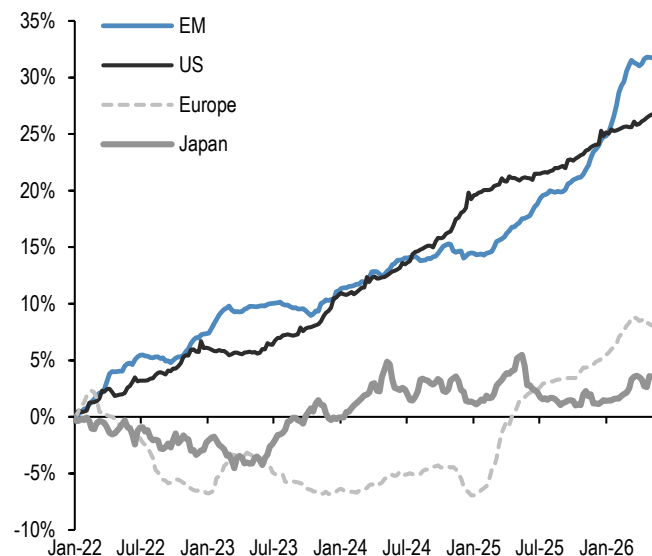
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A5: Global Equity ETF Flows

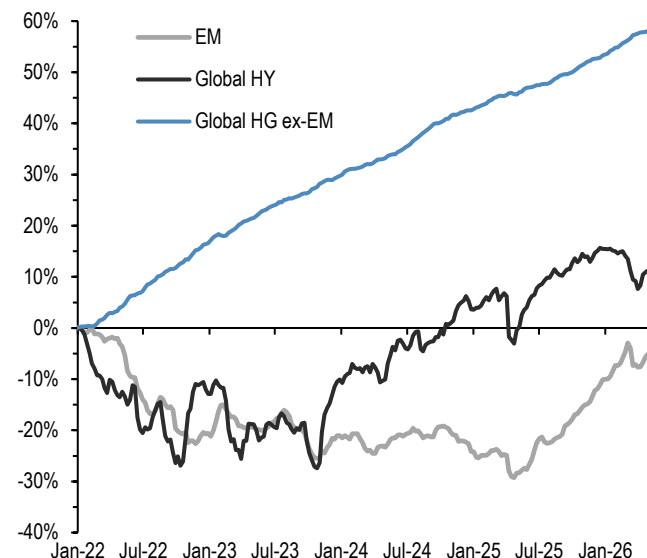
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
 Note: We include ETFs with AUM > \$200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

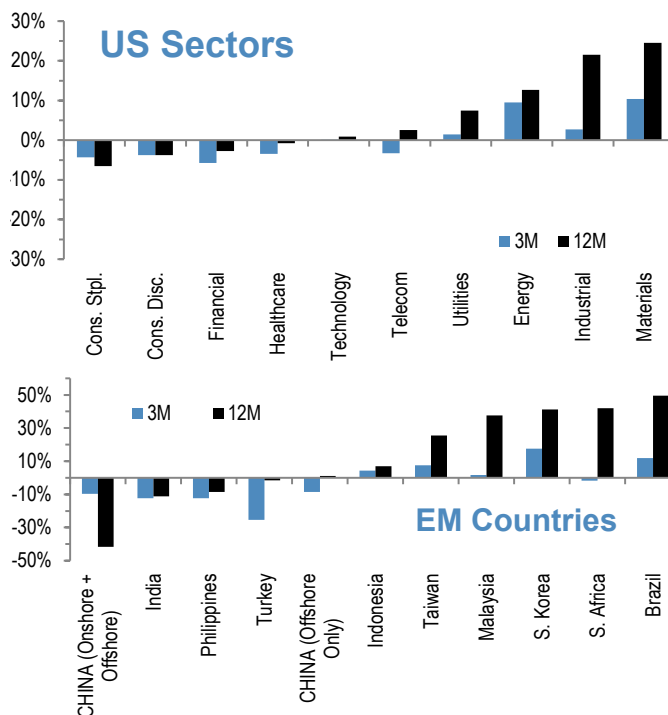
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A6a: Equity Sectoral and Regional ETF Flows

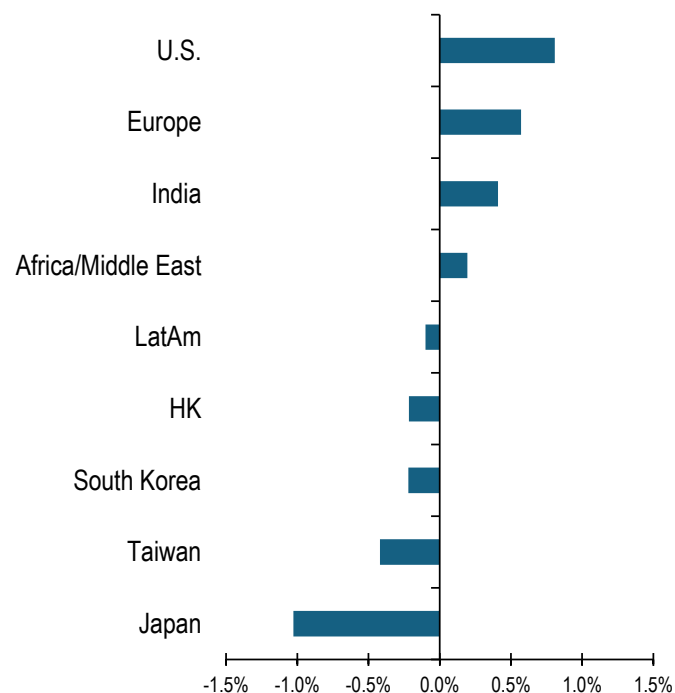
Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.



Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.

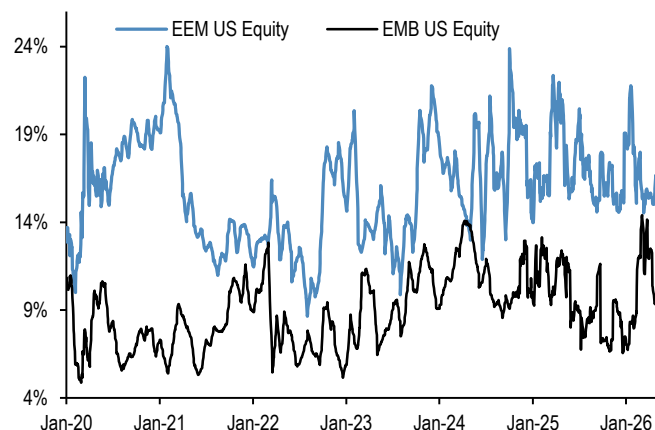
Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

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Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

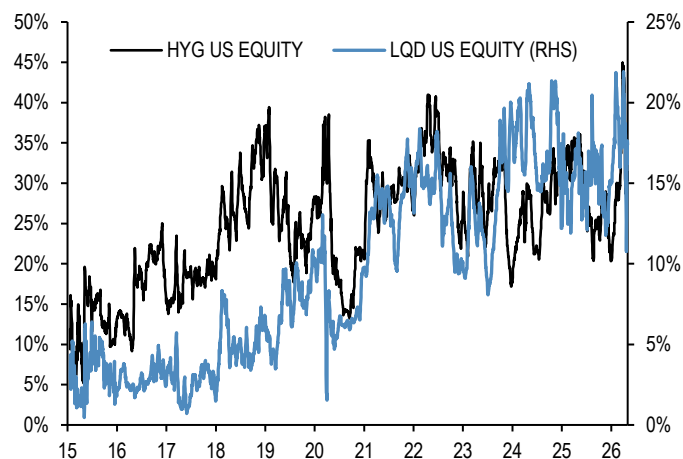
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

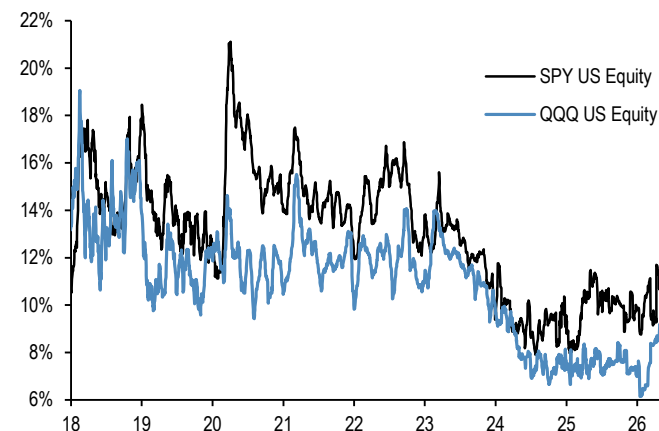
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

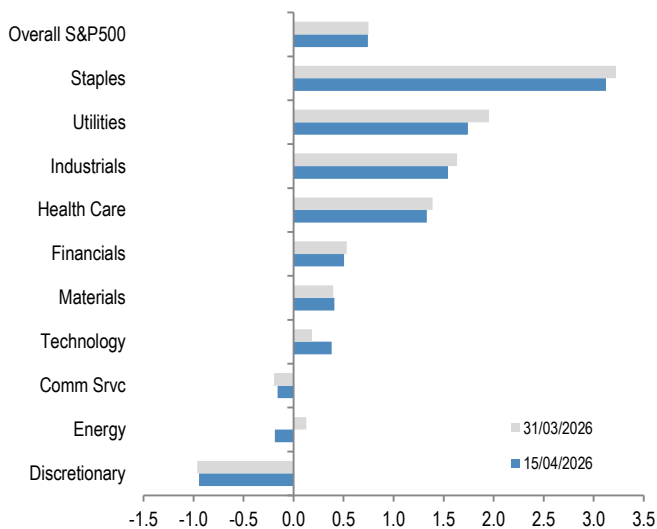
Short Interest as a % share of shares outstanding. Last obs is for 5th May 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28, 2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 28th Apr-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date		Upside	Downside	Implied/realized volatility
S&P 500	16%	13%	28-Aug-25	24%	27-Mar-26		8%	3%	1.27x
EuroSTOXX	20%	13%	18-Sep-25	26%	27-Mar-26		6%	7%	0.87x
Nikkei 225	26%	18%	22-Jul-25	33%	31-Mar-26		7%	8%	1.01x
Hang Seng	20%	17%	23-Dec-25	24%	26-Mar-26		4%	2%	1.03x
MSCI EM	24%	12%	15-Aug-25	35%	27-Mar-26		10%	12%	0.92x
Gold	23%	14%	19-Aug-25	33%	26-Mar-26		11%	9%	1.09x
Oil (brent)	64%	27%	19-Sep-25	89%	30-Mar-26		25%	37%	0.72x
Copper	23%	14%	18-Aug-25	37%	03-Feb-26		13%	9%	0.96x
BB commodity index	23%	13%	10-Dec-25	25%	30-Mar-26		2%	10%	0.98x
EUR/USD	9%	9%	03-Oct-25	14%	06-May-25		4%	0%	1.59x
USD/NOK	9%	0%	04-May-26	12%	06-May-25		3%	9%	1.35x
USD/JPY	8%	0%	04-May-26	12%	06-May-25		4%	8%	0.78x
GBP/USD	7%	6%	12-Dec-25	9%	07-May-25		2%	1%	0.94x
USD/CHF	7%	0%	04-May-26	11%	06-May-25		4%	7%	0.80x
10y US swaps	7%	6%	13-May-25	10%	27-Mar-26		3%	1%	1.06x
10y Eur swaps	69%	43%	26-Feb-26	97%	27-Mar-26		27%	26%	0.80x
CDX IG	47%	38%	29-Jul-25	66%	27-Mar-26		19%	9%	1.06x
CDX HY	35%	31%	09-Jan-26	57%	30-Mar-26		22%	4%	0.92x
iTraxx	49%	0%	04-May-26	70%	27-Mar-26		21%	49%	0.95x
iTraxx X/O	21%	19%	23-Dec-25	26%	07-May-25		5%	2%	0.43x

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Note: Swaps volatility is 3m 10y payer ATM implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

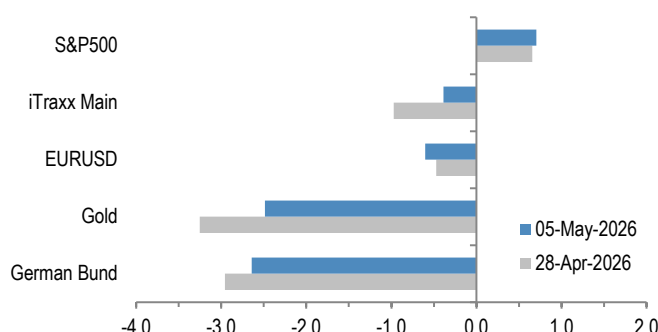
Definitions:

Current:	Latest available closing level (04-May-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

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Chart A11b: Option skew monitor

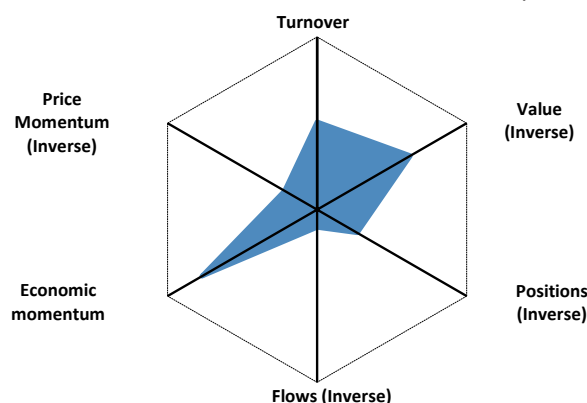
Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.



Source: J.P. Morgan Flows & Liquidity.

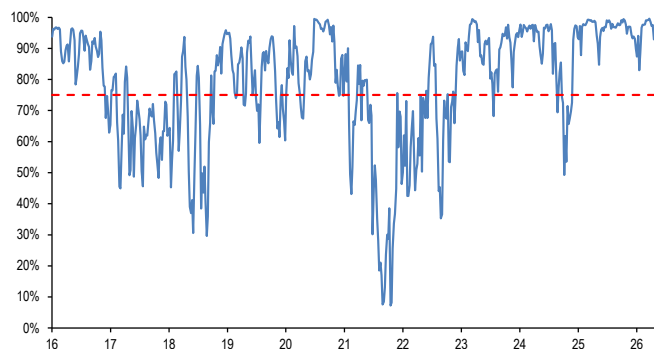
Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations. **Turnover:** Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value:** Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions:** Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows:** Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum:** Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum:** Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 1st May.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

In %.

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
 Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
 Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLows	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03



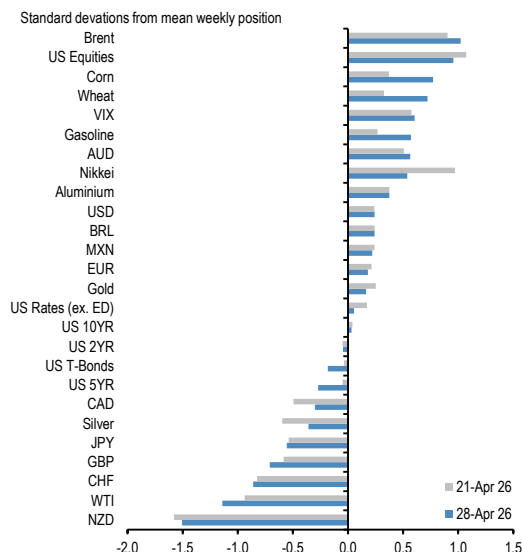
Source: J.P. Morgan Flows & Liquidity.

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Spec position monitor

Chart A12: Weekly Spec Position Monitor

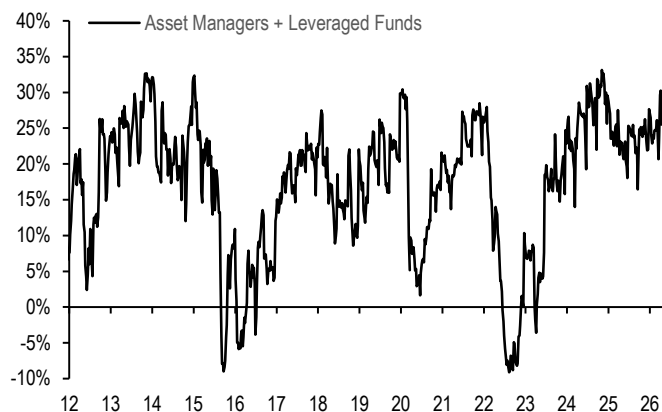
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

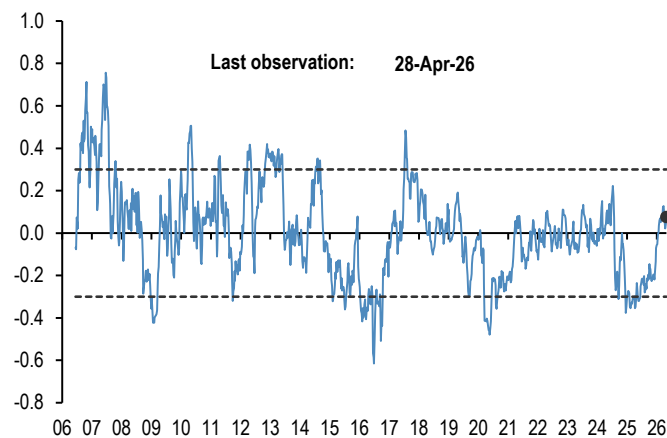
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

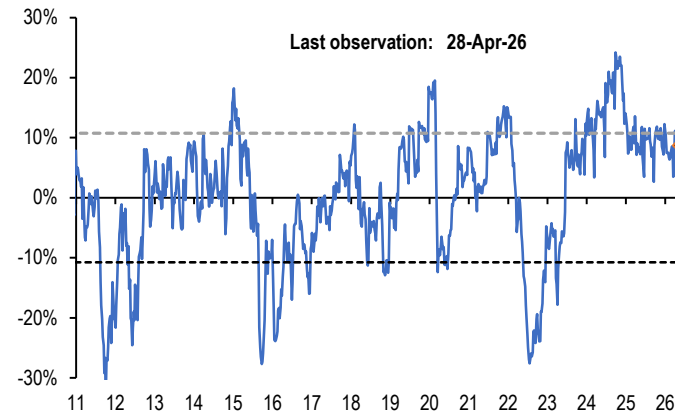
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the 'risky' and 'safe' series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD, CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.

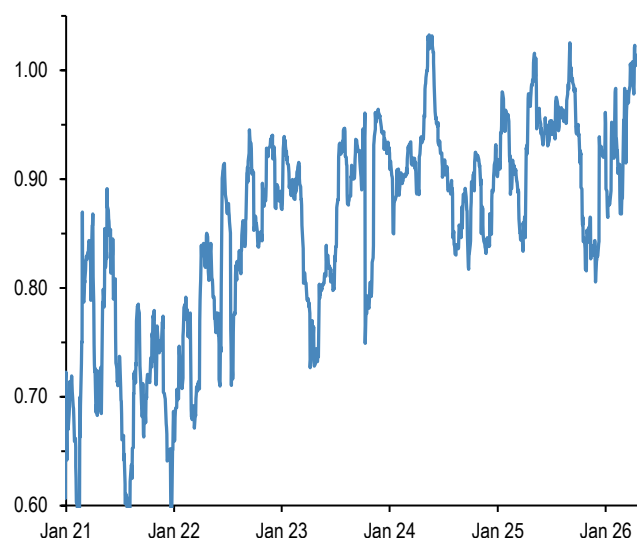


Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

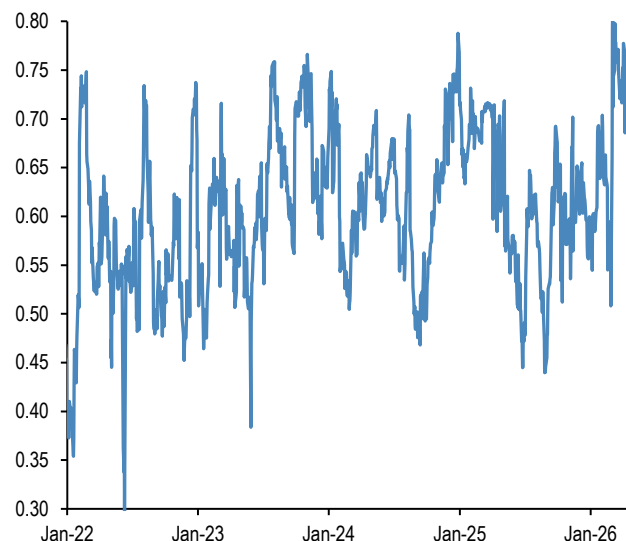
The table depicts the performance of various types of investors in % as of 4th May 2025.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	6.3%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	3.7%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	8.7%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	5.5%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	3.9%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	6.8%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	0.0%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	12.9%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	3.5%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

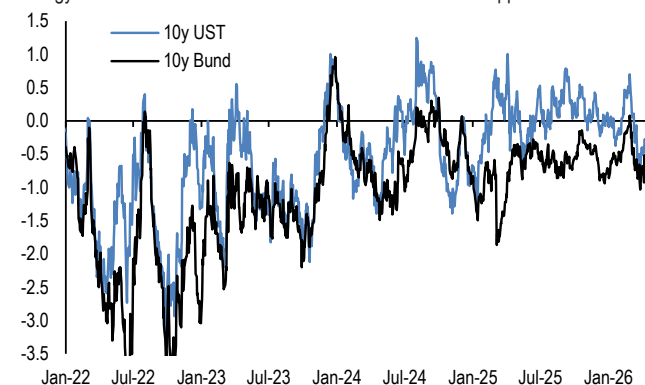
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.

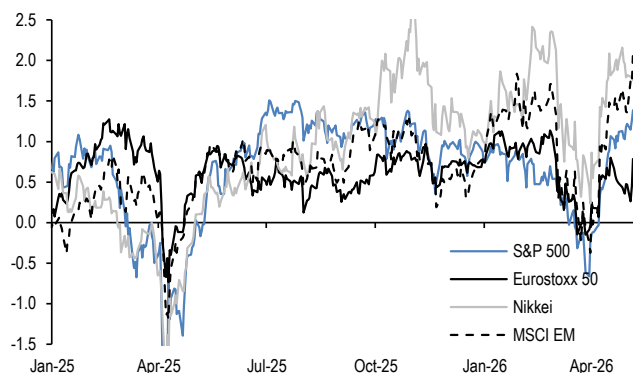


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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Chart A20: Momentum signals for S&P500

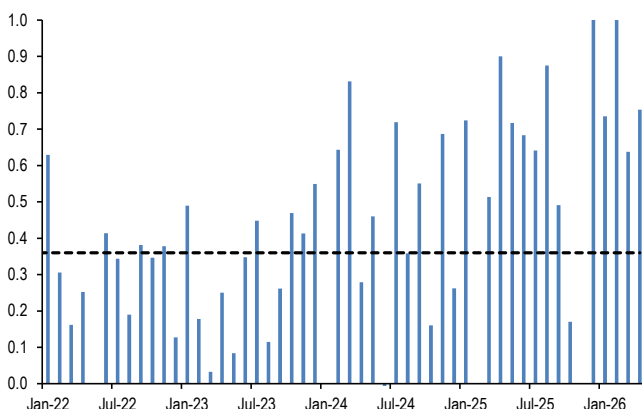
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

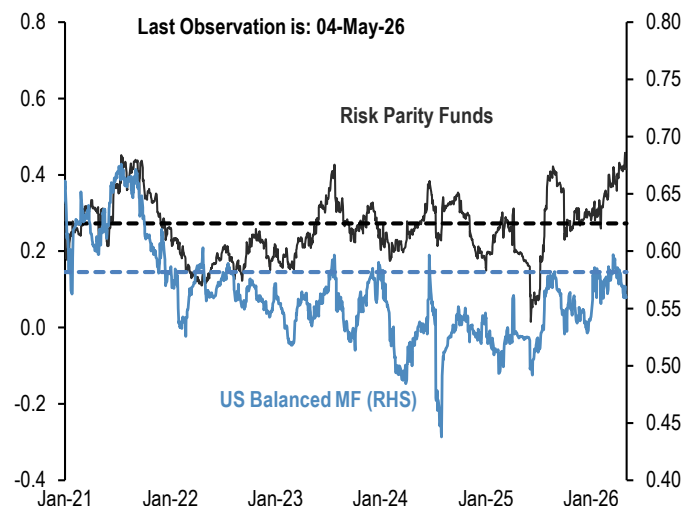
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is Apr 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

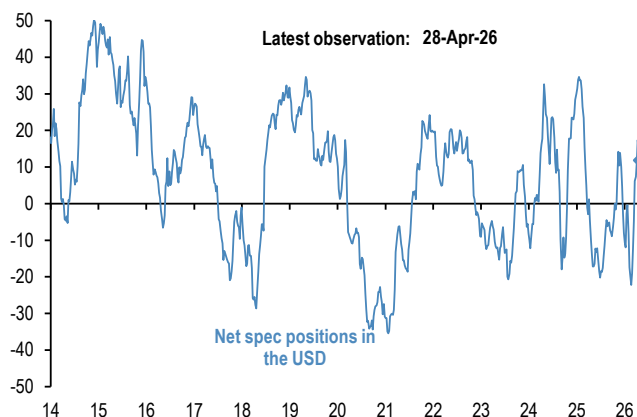
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of \$700bn, which is around half of the total AUM of \$1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry. It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21				1	0	1.1	6.9%
	long	462	5.5	22.7	0.24	0	11	2.4	70.3%
Brent	short	84				0	46	3.2	38.4%
	long	504	4.4	21.5	0.21	0	11	2.7	76.3%
Unleaded gas	short	105				0	44	3.5	47.6%
	long	483	2.6	23.2	0.11	0	38	2.8	74.8%
Heat Oil	short	63				0	11	2.3	23.7%
	long	483	4.3	21.9	0.19	0	45	3.4	101.9%
Gasoil	short	63				0	46	3.0	32.6%
	long	378	8.3	21.1	0.39	0	45	4.0	114.4%
Nat gas	short	105				-1	37	-0.6	-11.5%
	long	315	13.6	36.0	0.38	-1	66	-0.8	-25.4%
EU emission allowances	short	42				1	18	0.5	4.8%
	long	483	10.0	28.9	0.35	0	0	0.0	0.6%
1m fwd TTF Nat gas	short	21				1	4	0.5	4.6%
	long	294	36.5	31.3	1.17	1	31	0.8	36.6%
Gold	short	21				-1	6	-1.2	-3.3%
	long	483	3.1	10.6	0.29	0	306	2.1	26.7%
Silver	short	10				-1	3	-0.6	-2.2%
	long	462	6.4	19.6	0.33	0	158	2.3	54.7%
Palladium	short	42				-1	1	-0.1	-1.2%
	long	273	12.2	23.0	0.53	1	27	0.3	6.6%
Platinum	short	105				-1	11	-0.9	-8.3%
	long	273	4.3	18.6	0.23	1	8	1.0	16.1%
Aluminium	short	105				0	0	1.6	12.3%
	long	357	5.8	15.2	0.38	0	26	2.0	29.1%
Copper	short	147				1	58	0.6	6.9%
	long	399	7.2	17.2	0.42	1	64	1.1	23.7%
Lead	short	126				-1	70	-0.1	-1.6%
	long	357	0.7	19.3	0.04	-1	77	-0.2	-5.1%
Nickel	short	42				1	16	1.2	9.8%
	long	336	12.2	23.0	0.53	1	90	0.7	19.2%
Zinc	short	126				1	26	0.4	4.0%
	long	399	7.9	19.6	0.40	1	156	0.6	13.2%
Wheat	short	168				1	54	0.9	10.7%
	long	294	4.2	23.2	0.18	1	16	0.5	7.9%
Kansas wheat	short	147				0	5	1.5	18.5%
	long	483	8.3	20.8	0.40	1	13	0.4	9.3%
Corn	short	63				1	11	0.5	3.8%
	long	399	7.7	16.5	0.47	-1	0	-0.1	-1.3%
Soybeans	short	42				1	4	0.3	1.8%
	long	231	5.1	14.5	0.35	1	64	0.6	7.9%
Cotton	short	168				0	0	1.5	20.8%
	long	483	6.1	18.5	0.33	1	14	0.5	11.9%
Sugar	short	63				1	4	0.7	6.1%
	long	252	6.2	21.6	0.29	-1	26	-0.1	-1.2%
Coffee	short	63				0	0	0.0	0.0%
	long	273	7.8	23.8	0.33	-1	67	-0.5	-9.5%
Cocoa*	short	10				0	1	4.2	15.4%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.6	12.3	0.62	1	18	1.2	5.9%
	long	315				1	133	1.3	13.9%
Nasdaq 100	short	84	9.8	16.2	0.60	1	18	1.4	10.9%
	long	462				1	269	1.3	25.3%
Nikkei	short	63	3.9	13.3	0.29	1	5	1.1	6.3%
	long	420				0	24	2.5	36.8%
FTSE 100	short	168	5.2	12.1	0.43	1	43	0.5	3.5%
	long	504				0	26	1.5	17.2%
Eurostoxx 50	short	168	4.8	16.6	0.29	1	19	0.3	2.8%
	long	294				1	26	0.6	7.0%
MSCI EM	short	42	13.0	11.5	1.12	0	1	1.6	8.3%
	long	357				0	15	1.8	28.7%
HYG credit	short	168	0.4	3.4	0.12	0	49	0.0	-0.1%
	long	273				1	0	0.1	0.4%
LQD credit	short	10	2.7	6.5	0.41	-1	0	-0.2	-0.2%
	long	189				-1	0	-0.2	-0.9%
2Y USTs	short	252	0.8	1.1	0.77	-1	46	-0.7	-0.9%
	long	462				-1	44	-0.5	-0.9%
5Y USTs	short	252	1.5	2.8	0.54	-1	39	-0.5	-1.3%
	long	420				-1	10	-0.3	-0.9%
10Y USTs	short	42	2.0	3.6	0.55	-1	10	-0.5	-0.7%
	long	441				-1	10	-0.2	-1.1%
2Y Schatz	short	189	0.6	0.8	0.81	-1	15	-1.4	-1.1%
	long	504				0	5	-1.6	-2.7%
5y Bobl	short	63	1.4	1.9	0.72	-1	19	-1.0	-0.9%
	long	483				-1	296	-1.0	-3.4%
10y Bund	short	105	2.2	3.4	0.64	-1	43	-0.6	-1.3%
	long	462				-1	645	-0.8	-4.3%
10Y JGB	short	126	0.9	2.3	0.37	0	6	-1.6	-1.7%
	long	273				0	43	-2.4	-3.5%
10Y Gilts	short	42	2.3	4.0	0.57	0	0	-1.5	-2.4%
	long	399				-1	172	-1.3	-6.8%
10y OAT vs Bund	short	189	0.4	1.5	0.29	1	26	0.7	0.6%
	long	294				1	47	0.8	0.8%
10y BTP vs. Bund	short	84	2.3	6.3	0.36	-1	11	-0.1	-0.4%
	long	273				1	3	0.1	0.5%
Euro	short	42	3.3	6.4	0.51	1	3	0.1	0.1%
	long	273				-1	46	-0.4	-2.4%
Yen	short	21	3.5	6.3	0.56	1	3	0.2	0.4%
	long	462				-1	3	-1.3	-11.0%
Sterling	short	168	1.3	7.1	0.18	0	0	0.0	-0.2%
	long	294				-1	48	-0.2	-1.1%
AUD	short	42	4.0	7.8	0.50	1	18	0.5	1.5%
	long	420				1	73	0.6	5.5%
CAD	short	168	1.3	5.9	0.22	-1	1	-0.1	-0.4%
	long	504				-1	283	-0.6	-3.9%
G10 FX	short	42	2.1	5.2	0.40	-1	16	-0.4	-0.7%
	long	189				0	1	0.0	0.0%
EMCI	short	42	2.6	5.0	0.53	1	11	0.8	1.5%
	long	189				1	212	0.9	3.7%

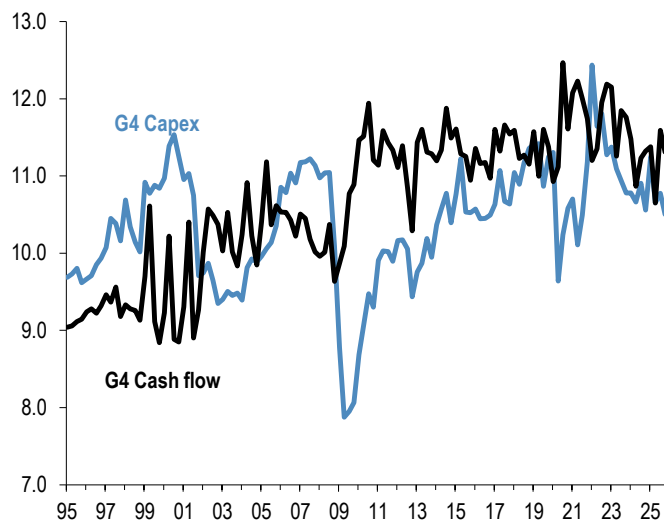
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

06 May 2026

Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

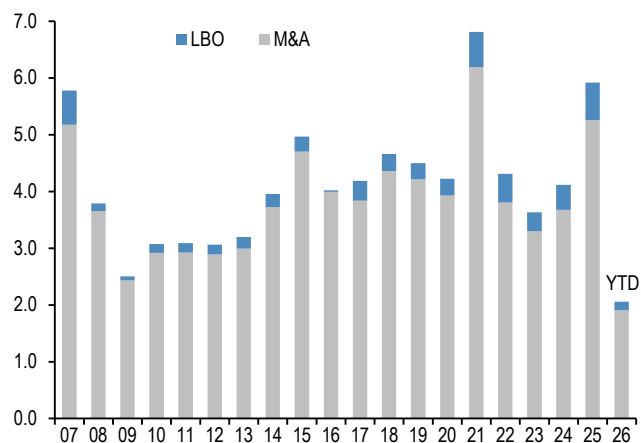
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

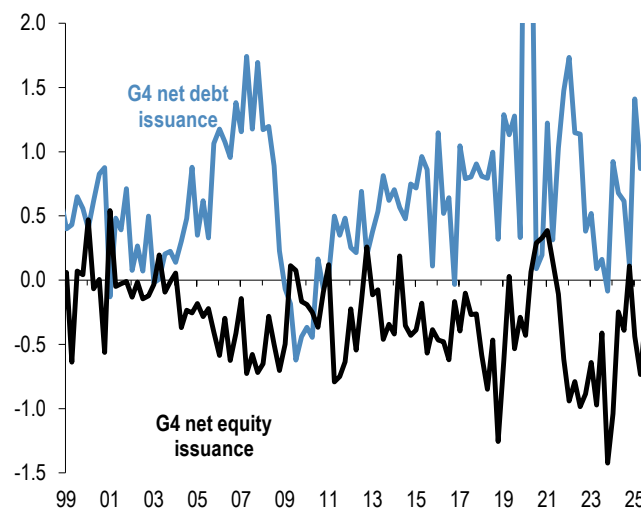
\$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

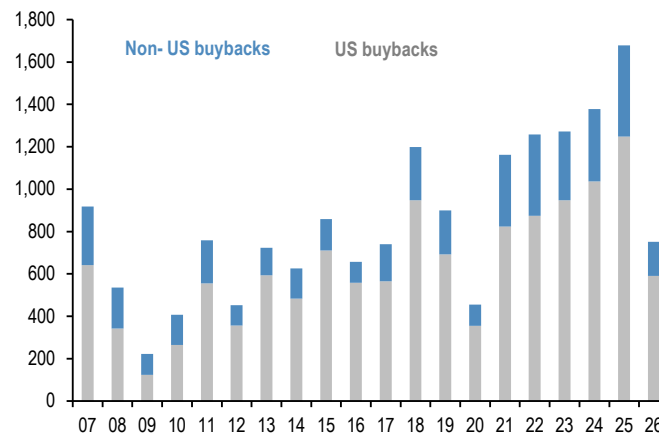
\$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

\$bn, are as of May 26. Buybacks are as announced.

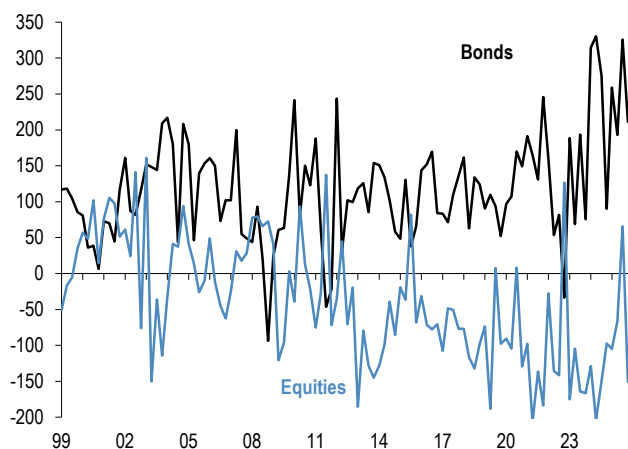


Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

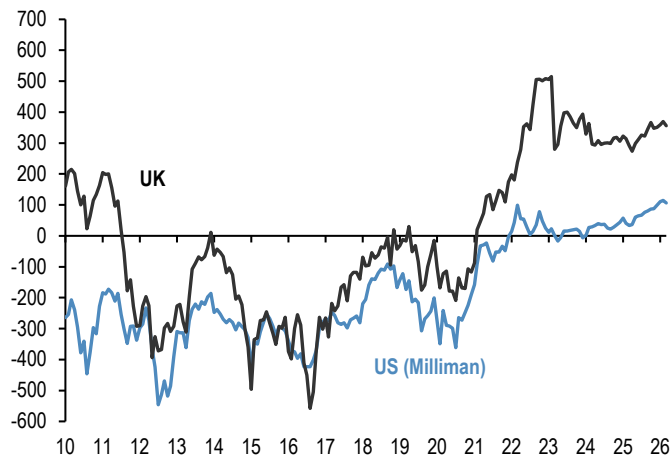
Equity and bond buying in \$bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

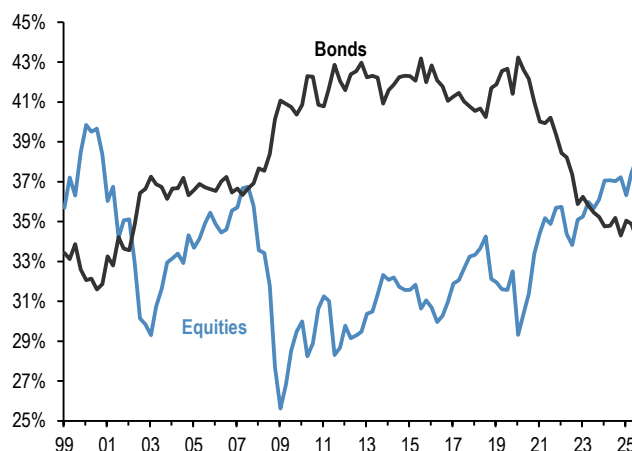
US\$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US\$ at today's exchange rates. Last obs. is Mar'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

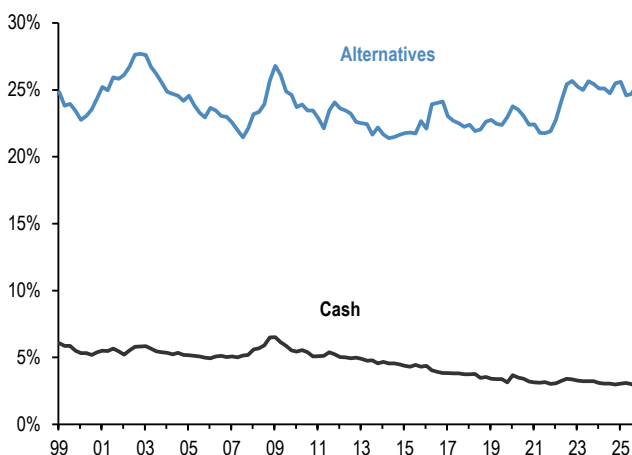
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q4 2025.



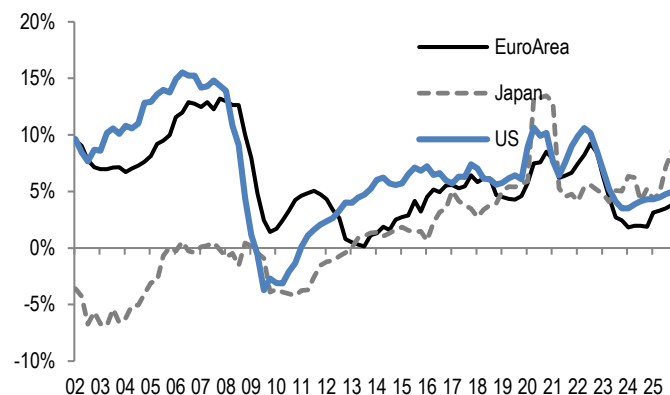
Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

06 May 2026

Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

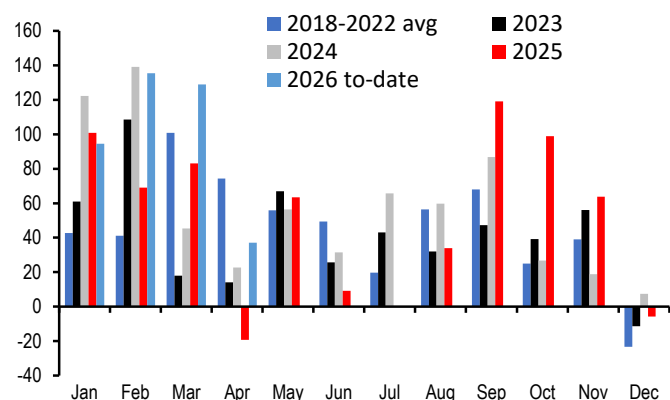
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q4'25 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

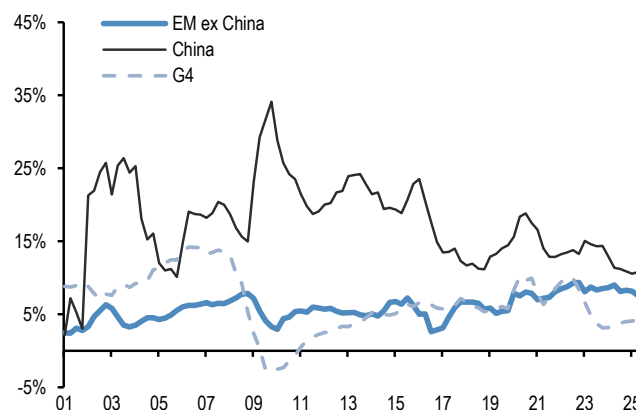
\$bn. Up to Apr 23rd.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q2'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

\$bn. Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

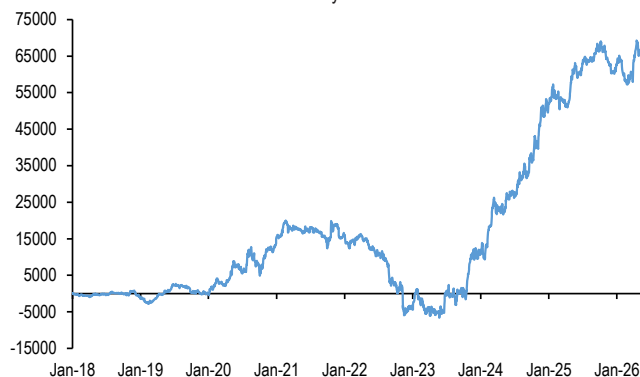
Equity Supply	24-Apr	4 wk avg	13 wk avg	y/y chng
Global IPOs	3.7	3.6	3.6	66%
Secondary Offerings	9.4	7.8	10.4	17%
Corporate announcements				
M&A - Global	97.8	122.2	120.3	29%
- US Target	58.6	55.2	57.3	49%
- Non-US Target	39.2	67.0	63.0	14%
Net bond issuance	Feb-26	3 mth avg	YTD avg	y/y chng
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

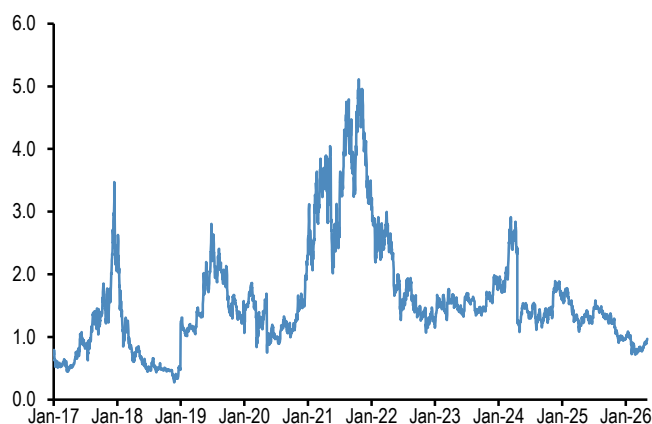
In number of contracts. Last obs. for 5th May 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

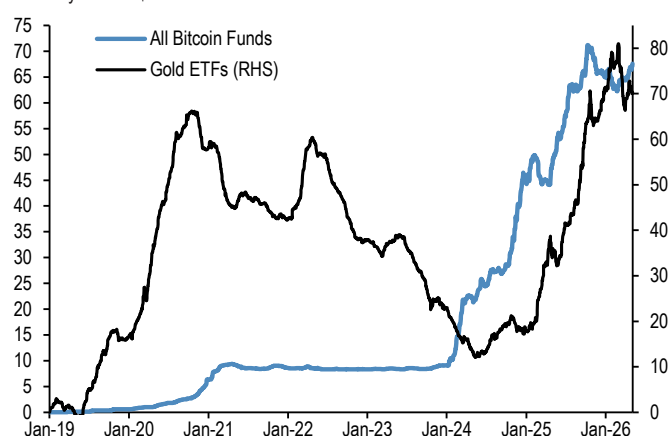
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

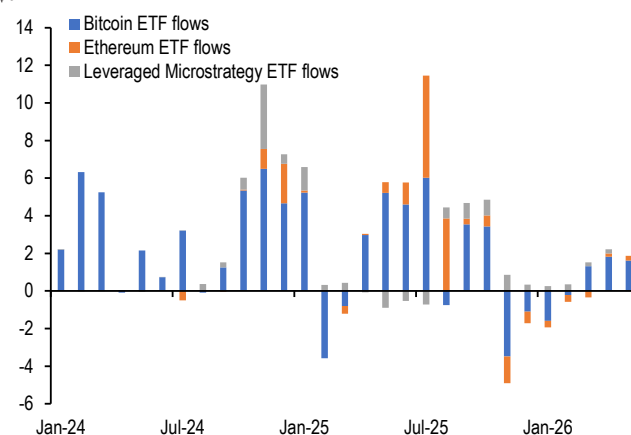
Both the y-axis in \$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

\$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

06 May 2026

Japanese flows and positions

Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

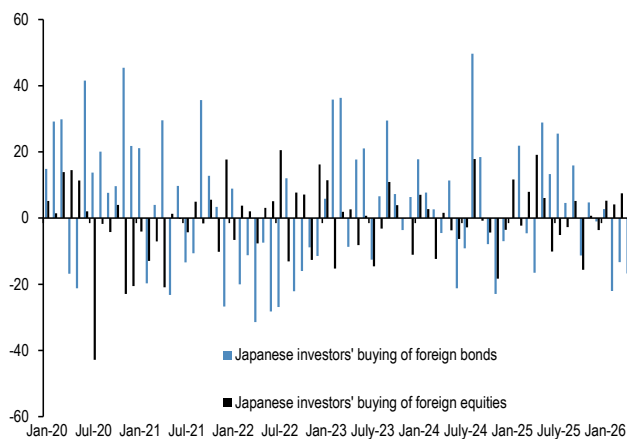
In bn of shares. Topix on right axis.



Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

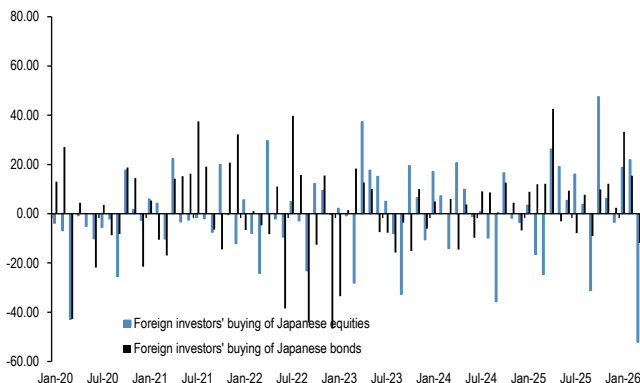
\$bn, Last weekly obs. is for 24th Apr' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

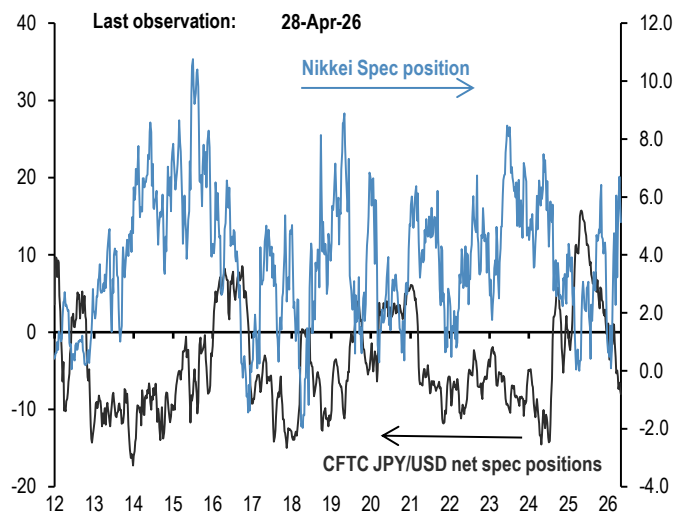
\$bn, Last weekly obs. is for 24th Apr' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in \$bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.

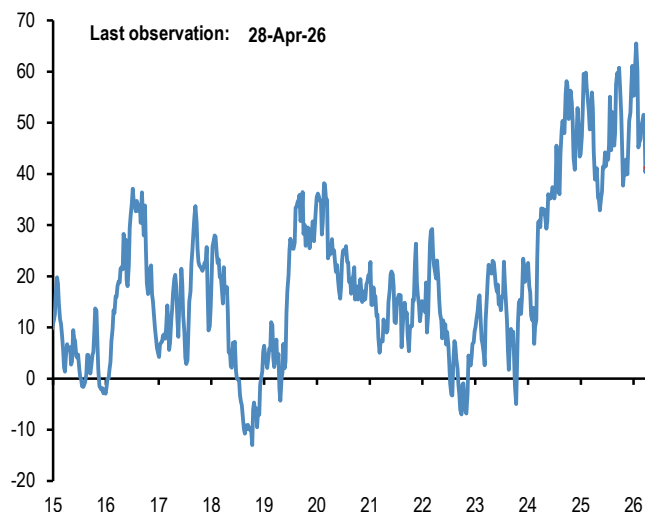


Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Commodity flows and positions

Chart A43: Gold spec positions

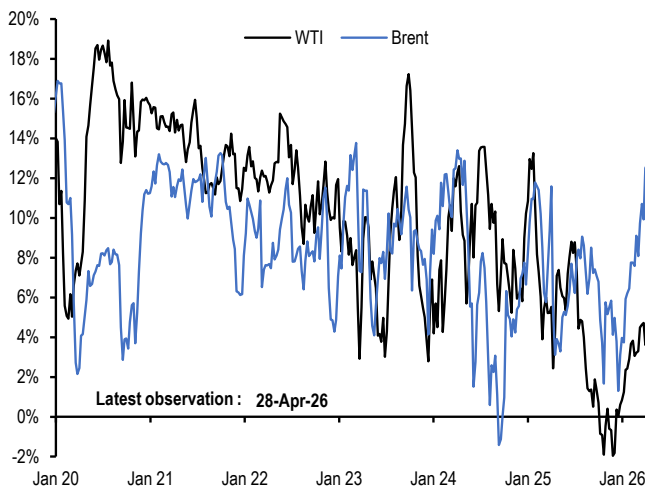
\$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

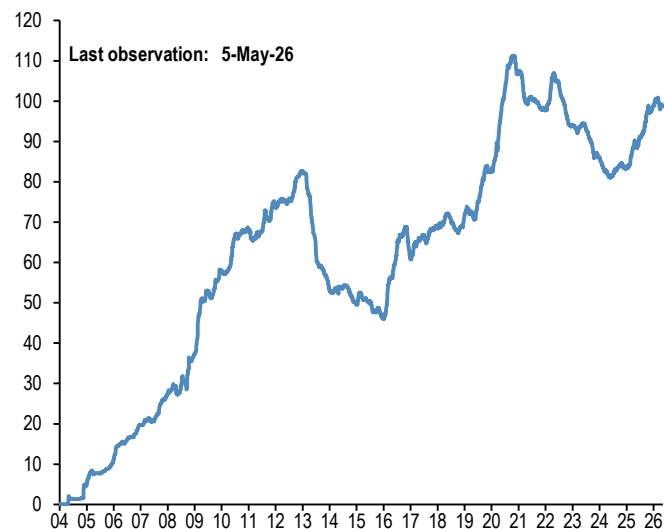
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

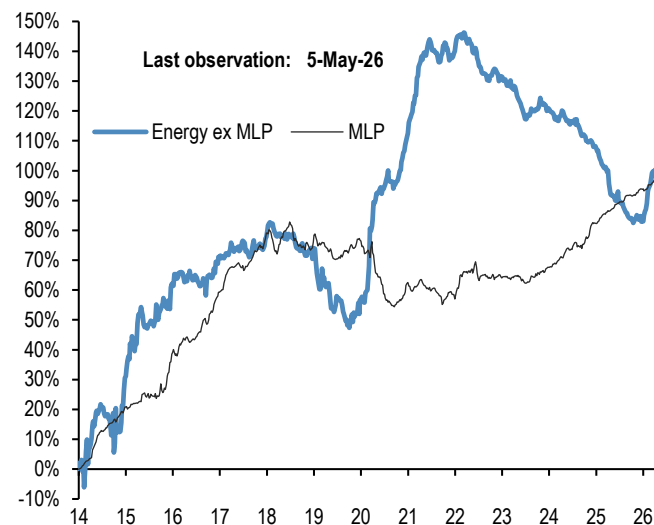
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.



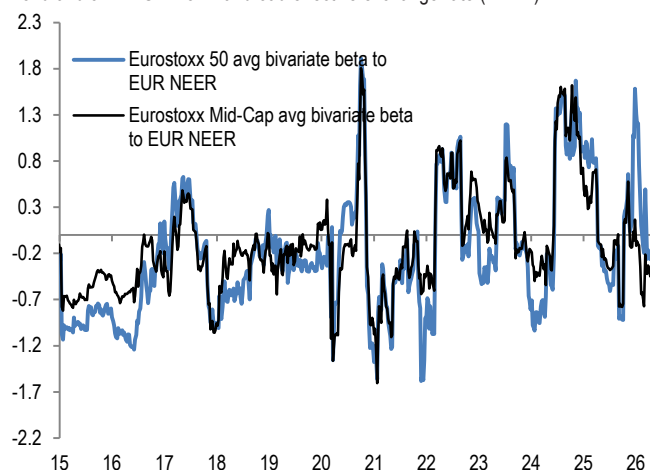
Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

06 May 2026

Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

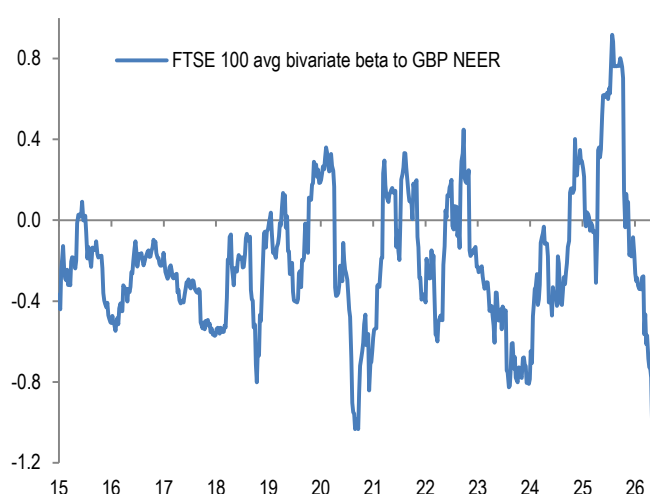
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

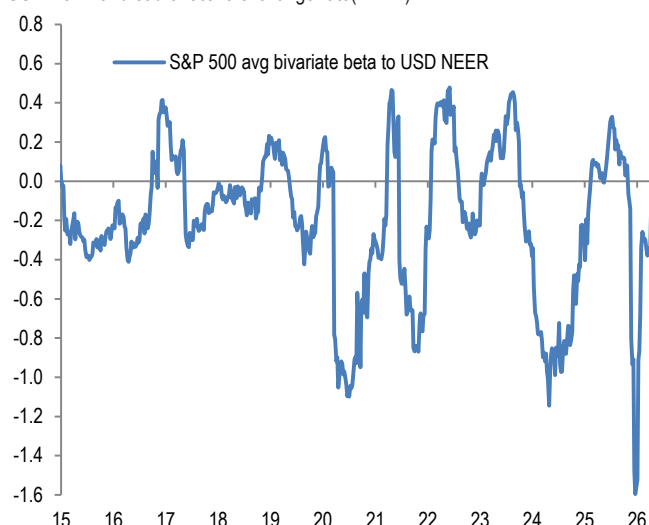
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

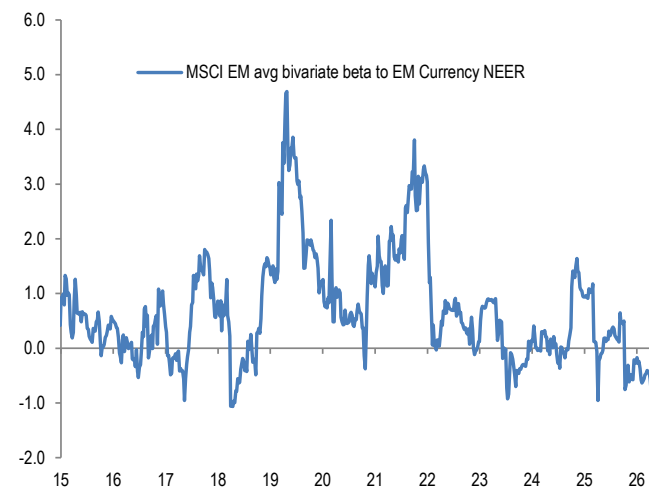
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).

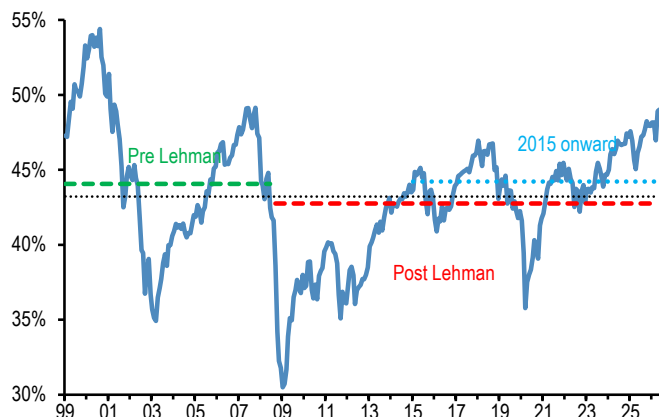


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

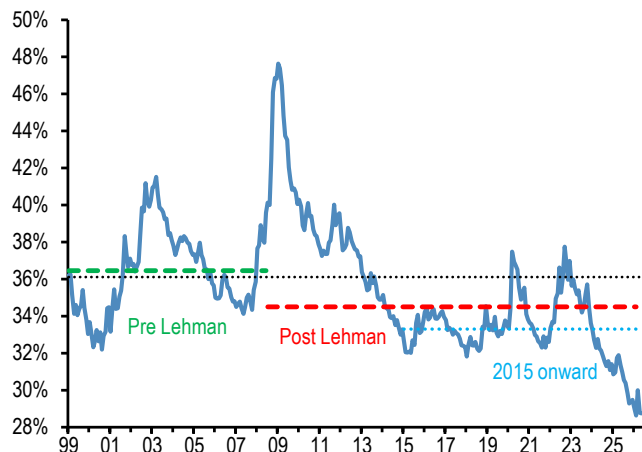
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

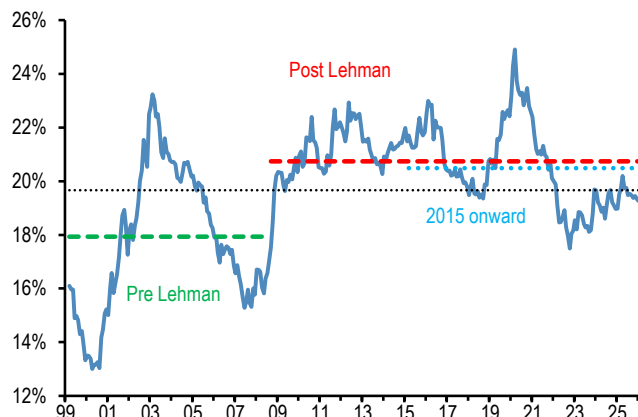
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

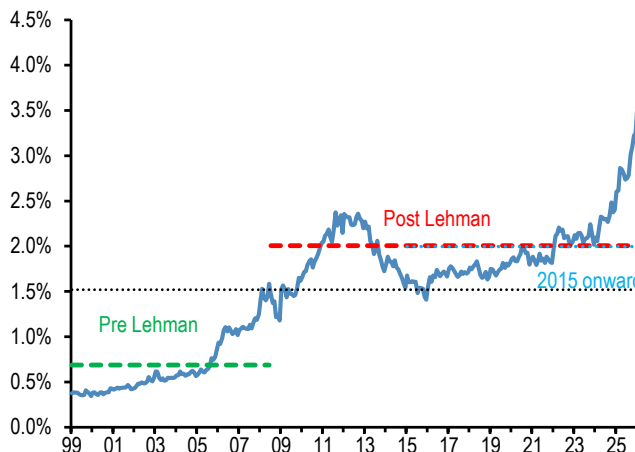
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

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Completed 06 May 2026 07:56 PM BST

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